STAT 24410/30030, Homework 6

Due Thursday, November 21, at 2:00pm

1. (20 pts) Lecture Notes, Chapter 5, Exercise 1.

2. (15 pts) Lecture Notes, Chapter 5, Exercise 2.

3. (15 pts) Lecture Notes, Chapter 5, Exercise 3.

4. (30 pts) Lecture Notes, Chapter 5, Exercise 4.

5. (20 pts) Here we use simulations to investigate power for one of the examples we investigated in class. Suppose, that $X_1, X_2, ..., X_{10}$ are iid Unif$(0, \theta)$. We would like to test $H_0 : \theta = 3$ against $H_A : \theta > 3$. Recall that we discussed two estimators for θ: (a) $\hat{\theta} = X_{(10)}$ (the largest observation); and (b) $\hat{\theta} = 2\bar{X}_{10}$. These suggest the following two tests with these two rejection regions:

$$S_{1,a} = \{X_{(10)} \leq A\} \cup \{X_{(10)} \geq 3\},$$

and

$$S_{1,b} = \{2\bar{X}_{10} \geq B\}.$$

(i) Use 10,000 simulations for $\theta = 3$ to determine A and B such that type 1 error is 0.05 for both tests. **Optional:** note that, if you want to verify your code/results, you can calculate A exactly (as we did in class) and you can find an approximation of B using CLT.

(ii) Using the values of A and B you found above, estimate the power of the two tests using simulations for θ =3.1, 3.2, and 3.3. Comment on which test you think is more powerful. **Optional:** you can also derive the power mathematically - exactly for (a) and approximately for (b).